Case 02 · Research and POC: Maturity Wall Simulator

Following the Magshimim "Research and POC" template: basic research, five key features, and the first POC (goals, plan, design, implementation and conclusions). Every POC was written in Python and actually run, and the numbers here are its output. Each POC states explicitly what is real and what is simulated.

Basic research: sequence of events

- **Load:** the system downloads a country's bond list from an official source (for each bond: amount, coupon, issue date, maturity date) together with the current yield curve.
- **Completeness check:** the sum of all bonds is compared with the marketable debt the country reports. A gap above a threshold stops the load and flags it.
- **Build the wall:** bonds are grouped by maturity year. Each brick in the wall is a bond with its coupon.
- **Run a scenario:** the user picks a future yield path. Each year the engine retires the maturing bonds, issues new ones in their place at that year's market yield, adds the primary deficit, and computes interest payments and debt.
- **Result:** a year-by-year path of interest payments and debt-to-GDP, with the year the effective rate overtakes growth marked.

Basic research: system components

- **Source Adapters:** one adapter per debt office, converting its format (API, CSV, Excel) into a uniform structure.
- **Portfolio Store:** a versioned store of bond lists (DuckDB), so loads can be compared over time.
- **Rollover Engine:** the core. Bond-by-bond rollover under a scenario.
- **Debt Dynamics:** the stock-flow debt equation and the r versus g engine.
- **Scenario API and Wall View:** a serving layer and the interactive wall view.

Basic research: reliability challenges

- **The rollover assumption:** at what tenor will a bond be issued to replace a maturing one? The debt office decides this every year. We assume the issuance mix is preserved and expose the assumption as a parameter.
- **Inflation-linked bonds:** their coupon changes with inflation. Not included in the POC.
- **Data quality:** if an adapter misreads a row, the whole projection shifts. The completeness check is the first line of defence.

Basic research: comparison with the leading competitor (IMF DSA)

- **Granularity:** the IMF works on aggregate debt with an approximate maturity profile. We work bond by bond.

- **Frequency:** the IMF updates as part of an annual country report. We update with every debt-office publication.
- **Transparency:** the IMF's DSA template is open, but needs an expert. Here every assumption is a visible slider.
- **Economic depth:** the IMF is far ahead. It includes growth and inflation scenarios and stress tests we do not have.

Conclusion of the comparison: our advantage lies in granularity and accessibility, not in the economics. The POC must test whether that granularity changes the answer, or whether the aggregate approximation is good enough.

Basic research: ethical issue

A public debt projection for a real country can move markets, especially under stress. A tool showing "France will reach 160% debt" may be read as a prophecy rather than a scenario. Response: every result is shown as a conditional scenario ("if yields stay like this"), with its assumptions beside it, and with no score or rating for the country.

Five key features

- **Bond-list loading:** Input: a debt-office file or API. Output: a uniform list of Bond objects. How: an adapter per source, normalised dates and currencies, a completeness check against reported debt.
- **Rollover engine:** Input: Bond list, yield path, issuance mix, primary deficit. Output: an annual path of interest and debt. How: a yearly loop that retires maturing bonds and issues new ones at the market yield.
- **r versus g engine:** Input: the effective-rate path and a growth forecast. Output: the year r overtakes g, and the primary surplus needed to stabilise. How: the stock-flow equation.
- **Maturity wall:** Input: Bond list and scenario. Output: a brick chart by year, coloured by coupon jump. How: D3, updating live with the slider.
- **Method comparison:** Input: the same scenario. Output: three paths (bond by bond, frozen rate, instant jump) on one chart. How: running the three engines side by side.

The POC: research question

Does a bond-by-bond calculation give a meaningfully different answer from the usual aggregate approximations? If not, the project has no reason to exist.

The POC: goals

- Build a rollover engine that works bond by bond, and verify it has no "leaks" (final debt equals initial debt plus deficits).
- **Compare it with two common aggregate approximations:** "frozen rate" (all debt keeps paying today's effective rate) and "instant jump" (all debt moves to the market rate at once, The Economist's exercise).

- Measure the gap in interest payments over 2027 to 2036 under three yield scenarios.
- **Check the engine against a known number:** the primary surplus France would need according to The Economist (about 1.8% of GDP).

The POC: technologies and limits

- Pure Python, no external libraries, so it runs in the browser on Pyodide.
- **Important limit:** the bond list is synthetic. We built it by simulating 35 years of issuance for a France-like country: stylised historical deficits (including jumps in 2009 and 2020), a stylised 10-year yield path, and a fixed issuance mix from one-year to 30-year bonds. Result: 63 bonds, debt of 115.5% of GDP, an effective rate of 2.43%. This is not France's actual bond list.
- **Out of scope:** inflation-linked bonds, floating-rate debt, foreign currency, user interface.

The POC: flow

Synthetic bond list → completeness check → [bond-by-bond engine | "frozen" approximation | "jump" approximation] → the same yield and deficit scenario → comparison of interest and debt paths → report.

The POC: results

- **The maturity wall is uneven:** in 2027 to 2031 between 9% and 11% of GDP matures each year, after which the wall drops to 5.8% to 7.6%. Debt maturing by the end of 2028 carries an average coupon of only 2.06%, so almost all of it will be refinanced at a higher cost.
- **"Yields stay" scenario (3% on the 10-year):** in 2031 bond-by-bond interest payments are 3.82% of GDP. The frozen approximation says 3.07% (an underestimate of about three-quarters of a point of GDP), and the instant jump says 4.26%. In 2036 debt reaches 157% bond by bond, against 150% and 161% for the approximations.
- **"Up one point" scenario:** in 2036 interest reaches 6.3% of GDP and debt 164%. The frozen approximation, which is not sensitive to yields at all, keeps saying 150%.
- **The most surprising result, the "down one point" scenario:** the instant jump says interest payments will fall (from 2.75% to 2.56% of GDP in the first year). The bond-by-bond engine says they rise regardless, from 2.75% to 3.19% in 2031 and 3.6% in 2036. The reason: the old debt carries coupons even lower than the new market rate. Even if yields fall, the bill still goes up.
- **Check against a known number:** by the formula, if all debt is refinanced at 3.5% with 2% nominal growth, a primary surplus of 1.7% of GDP is needed to stabilise the debt. The Economist arrived at about 1.8% for the real France. Same order of magnitude, which indicates the engine behaves correctly.

The POC: what is real and what is simulated

Real: the rollover mechanism, the debt equation, and the comparison of the three methods on the same input. Simulated: the bond list, the historical yield path, and growth. So the exact numbers

say nothing about France. What can be concluded: the gap between methods is large enough to change a conclusion (up to three-quarters of a point of GDP a year), and the falling-yield scenario shows that a poor approximation can even get the direction wrong.

The POC: conclusions

✅ Concept proven. Bond-level detail changes the answer, not just its precision. But the comparison is only partly fair: we compared against two simple approximations, not against the full IMF DSA model, which already accounts for the maturity profile approximately. The open question for the spec: how much of the advantage we found survives against professional tools, not just against a thought experiment.